

Induction Continued, Strong Induction

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Overview

- 1 Induction from a Starting Point
- 2 Strong Induction (5.2)
- 3 Optional topics

Induction starting at a value other than 0

Our previous recipe for induction: to show $P(n)$ for every $n \in \mathbb{N}$, it suffices to show

- $P(0)$
- For an arbitrary n , if $P(n)$ holds, then $P(n + 1)$ holds.

Claim: to show $Q(n)$ for every $n \geq k$, it suffices to show

- $Q(k)$
- for an arbitrary $n \geq k$, if $Q(n)$ holds, then $Q(n + 1)$ holds.

Proof. Suppose we have a predicate $Q(n)$ with those properties. Define $P(n) = Q(n + k)$. We show $\forall n, P(n)$ holds by induction. $P(0) = Q(0 + k)$ holds, by assumption 1. Suppose $P(n)$ holds for an arbitrary n . This means $Q(n + k)$ holds. Since $n + k \geq k$, assumption 2 says $Q(n + k + 1)$ holds. And this is $P(n + 1)$. So by induction, we have shown $P(n) = Q(n + k)$ holds for all n : that is, $Q(n)$ holds for all $n \geq k$.

Example

Theorem. For every natural number $n \geq 5$, $2^n > n^2$.

Proof. By induction on n starting at 5, with predicate $P(n)$ defined to be $2^n > n^2$.

Base case: $2^5 = 32 > 25 = 5^2$.

Inductive step: suppose $n \geq 5$ and $2^n > n^2$. We want to show $2^{n+1} > (n+1)^2$. Since $n \geq 5$, we have $2n+1 \leq 3n \leq n^2$. So:

$$\begin{aligned}(n+1)^2 &= n^2 + 2n + 1 \\ &\leq n^2 + n^2 \\ &< 2^n + 2^n \\ &= 2^{n+1}\end{aligned}$$

This concludes our proof by induction from a starting point.

Strong Induction

Compare these two different “induction step” rules:

Ordinary induction: for all $n \in \mathbb{N}$, $P(n)$ implies $P(n + 1)$.

Strong induction: for all $n \in \mathbb{N}$, $P(0), P(1), \dots, P(n)$ together imply $P(n + 1)$.

Strong induction lets you assume more in your inductive step. If you plan to use it in a proof, make sure you say so up front.

A recipe

To prove by strong induction that $P(n)$ holds for every $n \geq k$, we need to show that

- $P(k)$ holds.
- For any $n \geq k$, if $P(k), \dots, P(n)$ all hold, then $P(n + 1)$ holds.

Combines strong induction and induction from a starting point! What happens when $k = 0$?

Example usage of strong induction

Theorem: Every integer greater than 1 is a product of primes.

Our proof will use strong induction on $P(n)$ “ n is a product of primes”.

We need to prove that $P(n)$ holds for all $n \geq 2$.

Base Case ($n = 2$): $P(2)$ is true because 2 is prime, so it is the product of (one) primes.

Inductive step

Inductive step: Suppose that $n \geq 2$ and that k is a product of primes for every integer k where $2 \leq k \leq n$. We must show that $P(n + 1)$, that is, $n + 1$ is a product of primes. We proceed by cases:

- $n + 1$ is prime: It is the product of (one) primes, so $P(n + 1)$ holds.
- $n + 1$ is not prime: By definition, $n + 1 = km$ for some integers k, m such that $2 \leq k, m \leq n$. By the strong induction hypothesis, both k and m are products of primes. So, therefore, so is km and so is $n + 1$. Again, $P(n + 1)$ holds.

That completes the cases. And, that completes the strong induction proof.

Coins of 3 and 5 units

What values can you make if you have coins worth 3 and 5 units?

- 1? No.
- 2? No.
- 3? Yes, one 3-unit coin: $(0, 1)$.
- 4? No.
- 5? Yes, one 5-unit coin: $(1, 0)$.
- 6? Yes, two 3-unit coins: $(0, 2)$.
- 7? No.
- 8? Yes: $(1, 1)$.
- 9? Yes: $(0, 3)$.
- 10? Yes: $(2, 0)$.
- 11? Yes: $(1, 2)$.

Coin Theorem

Theorem: You can make all values 8 or larger using coins of value 3 and 5.

Proof: We proceed by strong induction with induction hypothesis $P(n)$: There is a collection of coins whose value is n .

Base case: $P(8)$ is true because we can make 8 using $(1, 1)$.

Inductive step: Assume $P(k)$ holds for all integers $8 \leq k \leq n$, and now prove $P(n + 1)$ holds. We argue by cases:

- $n = 9$: $(0, 3)$
- $n = 10$: $(2, 0)$
- $n \geq 11$: By strong induction, we can make the value $n - 2$, then add a 3-unit coin to get $n + 1$.

That completes the proof by strong induction, which proves the theorem. QED.

The game

Start with a list of numbers and a score of zero. Pick an integer z from the list and replace it with two positive integers x and y such that $x + y = z$. Add xy to your score. Continue until list consists of all 1s. Try to maximize your score.

Example:	7							0
	5				2			10
	2		3		2			16
	2		2		1	2		18
	1	1	2		1	2		19
	1	1	2		1	1	1	20
	1	1	1	1	1	1	1	21

Irrelevance theorem

Theorem: No matter how you play the game starting from $n \geq 1$, your score will be $n(n-1)/2$.

Proof: The proof is by strong induction with $P(n)$ as the proposition that every way of playing starting with the number n gives a score of $n(n-1)/2$.

Base case: Starting from a list of $n = 1$, the game ends immediately with a score of $n(n-1)/2 = 1(0)/2 = 0$. $P(1)$ is true.

Inductive step: We must show that $P(1), \dots, P(n)$ imply $P(n+1)$ for all $n \geq 1$. Assume that $P(1), \dots, P(n)$ are all true and we are playing starting with $n+1$. The first move must break $n+1$ into positive numbers a and b with $a+b = n+1$. The total score for the game is the sum of points for this first move plus the scores obtained from playing with a and b :

Inductive step

$$\begin{aligned}
 \text{total score} &= \text{score for turning } n + 1 \text{ into } a \text{ and } b \\
 &\quad + \text{score from starting with } a \\
 &\quad + \text{score from starting with } b && \text{rules of game} \\
 &= ab + \frac{a(a-1)}{2} + \frac{b(b-1)}{2} && \text{inductive hyp.} \\
 &= \frac{2ab + a^2 - a + b^2 - b}{2} \\
 &= \frac{(a+b)^2 - (a+b)}{2} && \text{algebra} \\
 &= \frac{(n+1)^2 - (n+1)}{2} && \text{Defn } a, b \\
 &= (n+1)n/2 && \text{algebra}
 \end{aligned}$$

We showed $P(1), P(2), \dots, P(n)$ implies $P(n+1)$, showing the claim true by strong induction. QED.

Every number is interesting

Theorem. Every natural number is interesting.

Proof. We proceed by strong induction with the property $P(n) = “n \text{ is interesting.}”$

Base case: 0 is certainly interesting!

Inductive case: suppose $0, \dots, n$ are all interesting. If $n + 1$ were not interesting, it would be the smallest noninteresting number. But that's interesting!

Another version of strong induction

We can phrase the strong induction principle differently:

To show $P(n)$ holds for all $n \in \mathbb{N}$, it suffices to show that for an arbitrary n , if $P(k)$ holds for every $k < n$, then $P(n)$ holds.

No base case! Why not?

Proving strong induction

We can *prove* that this principle of strong induction works, using regular induction.

Suppose: for all $n \in \mathbb{N}$, if $P(k)$ holds for every $k < n$, then $P(n)$ holds. (Call this hypothesis H .)

We'll proceed by induction on the property $Q(n) = "P(k) \text{ holds for every } k < n."$

Base case: $Q(0)$ holds, since $P(k)$ holds for every $k < 0$.

Inductive step: Suppose $Q(n)$ holds. So $P(k)$ holds for every $k < n$. By our hypothesis H , $P(n)$ holds. Thus, $P(k)$ holds for every $k < n + 1$, which is $Q(n + 1)$.

Strong Induction vs. Induction vs. Well Ordering (5.3)

Covered in the book, Chapter 2.

Idea: If a claim has a counterexample, it has a *least* counterexample. And, if we can prove (by contradiction) that its existence implies an even smaller counterexample, then it wasn't the smallest counterexample after all. Indeed, there must be no counterexample (!).

It is basically a rephrasing of strong induction because we're showing that being true up to some value n means it can't be false for n .

And, strong induction is just ordinary induction with a universal quantifier in its key proposition.

So, three different names for the same thing. Use is a matter of taste.